

$$\mathcal{L} = i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi \quad \left| \begin{array}{l} (\sigma^\mu)_{\alpha\dot{\alpha}} = (1, \sigma^i)_{\alpha\dot{\alpha}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\beta} = (1, -\sigma^i)^{\dot{\alpha}\beta} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} = \epsilon^{\alpha\beta}\epsilon^{\dot{\alpha}\dot{\beta}}(\sigma^\mu)_{\beta\dot{\beta}} \\ (\sigma^\mu)_{\alpha\dot{\alpha}}(\sigma_\mu)_{\beta\dot{\beta}} = -2\epsilon_{\alpha\beta}\epsilon_{\dot{\alpha}\dot{\beta}} \end{array} \right. \quad \left| \begin{array}{l} \gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\alpha}} \\ \bar{\sigma}^{\mu\dot{\alpha}\alpha} & 0 \end{pmatrix} \\ \{\gamma^\mu, \gamma^\nu\} = -2\eta^{\mu\nu} \end{array} \right.$$

$$* (-i\not{\partial} + m)\Psi = 0 \quad \Rightarrow \quad (i\not{\partial} + m)(-i\not{\partial} + m)\Psi = 0$$

$$(-\partial^2 + m^2)\Psi = 0$$

$$\Psi(x) \sim u(p) e^{ipx} + v(p) e^{-ipx} \quad | \quad p^2 = p_\mu p^\mu = -m^2$$

$$(-\partial^2 + m^2)\Psi = 0 = (\partial_t^2 - \vec{\partial}^2 + m^2)\Psi = 0$$

$$(\partial_t^2 + \vec{p}^2 + m^2)\Psi = 0 \quad \Rightarrow \quad \partial_t^2 \hat{\Psi} = -(\vec{p}^2 + m^2)\hat{\Psi}$$

$$\Psi(p, t) = u(p) e^{-iEt} + v(p) e^{iEt} \quad E^2 = \vec{p}^2 + m^2$$

$$\Psi(x, t) = \int \frac{d^3p}{(2\pi)^3} e^{ipx} (u(p) e^{-iEt} + v(p) e^{iEt})$$

$$* (-i\not{\partial} + m)\Psi = \int \frac{d^3p}{(2\pi)^3} (\not{p} + m)u(p) e^{ipx} + (-\not{p} + m)v(p) e^{-ipx} = 0$$

so each must separately vanish since $e^{ix} e^{-ix}$ are linearly independent:

$$(\not{p} + m)u(p) = 0, \quad (-\not{p} + m)v(p) = 0$$

* Each has two independent solutions which we label

$$s = \pm$$

$$(\not{p} + m)u(p) = 0 = (\gamma^\mu p_\mu + m)u(p) = 0$$

$$u(p) = \int d^3x e^{-ipx} u(x) : \int d^3x (\gamma^\mu p_\mu + m) e^{-ipx} u(x) = 0$$

$$\left(\begin{pmatrix} 0 & p_\mu \sigma^\mu \\ p_\mu \bar{\sigma}^\mu & 0 \end{pmatrix} + m \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} u_1(p) \\ u_2(p) \end{pmatrix} = 0$$

$$= \begin{pmatrix} m & p_\mu \sigma^\mu \\ p_\mu \bar{\sigma}^\mu & m \end{pmatrix} \begin{pmatrix} u_1(p) \\ u_2(p) \end{pmatrix} = \begin{pmatrix} m u_1 + p_\mu \sigma^\mu u_2 \\ (p_\mu \bar{\sigma}^\mu) u_1 + m u_2 \end{pmatrix} = 0$$

$$u_1 + m^{-1}(p_\mu \sigma^\mu) u_2 = 0$$

$$(p_\mu \bar{\sigma}^\mu) u_1 + m u_2 = 0$$

$$(p_\mu \bar{\sigma}^\mu) (-m^{-1} p_\nu \sigma^\nu u_2) + m u_2 = 0$$

$$(p_\mu \bar{\sigma}^\mu)(p_\nu \sigma^\nu) = m^2$$

$$p_\mu p^\mu = -m^2$$

$$\frac{d^3p}{d\vec{p}} = \frac{d^3p}{(2\pi)^3 2E_p}$$

$$* \Psi(x) = \sum_{s=\pm} \int \frac{d^3p}{(2\pi)^3 2E_p} (b_s(p) u_s(p) e^{ipx} + d_s^\dagger(p) v(p) e^{-ipx})$$

Canonical Quantization: $(b_{\pm}^{\dagger}(p), d_{\pm}^{\dagger}(p))$: Creation
 $(b_{\pm}(p), d_{\pm}(p))$: Annihilation

$$b_{\pm}|0\rangle = d_{\pm}|0\rangle = 0$$

$$|p, \pm\rangle = b_{\pm}^{\dagger}|0\rangle$$

Spinor technology: $(\not{p} + m)u_s(\vec{p}) = 0$
 $(-\not{p} + m)v_s(\vec{p}) = 0$

Massive case: $p^{\mu} = (E, 0, 0, 0)$ Boost

Spin matrix: $S_z \equiv \frac{i}{4}[\gamma^1, \gamma^2] = \frac{i}{2}\gamma^1\gamma^2 = \begin{pmatrix} \frac{1}{2}\sigma_3 & 0 \\ 0 & \frac{1}{2}\sigma_3 \end{pmatrix}$
distinguish states as eigen vectors of the spin matrix

$$\gamma^{\mu} = \begin{pmatrix} 0 & \sigma^{\mu}_{ij} \\ \sigma^{\mu ic} & 0 \end{pmatrix}: \not{p} = \begin{pmatrix} 0 & -E\mathbb{1} \\ -E\mathbb{1} & 0 \end{pmatrix}$$

$$\text{But in } p^{\mu} = (E, 0, 0, 0) \Rightarrow E = m$$

$$\not{p} = -m\gamma^0 = -m \begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}$$

* We will require:

$$S_z u_{\pm}(\vec{0}) = \pm \frac{1}{2} u_{\pm}(\vec{0})$$

$$S_z v_{\pm}(\vec{0}) = \mp \frac{1}{2} v_{\pm}(\vec{0})$$